Detecting Price Anomalies and Benchmarking Variability in Public Sector Procurement Data

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Abstract*— *Purpose: *This project examines whether more meaningful price benchmarks and potential pricing anomalies can be identified in public sector procurement data after comparable purchases are normalized and grouped.*

Methods: *Using a de-identified set of Virginia procurement transactions obtained through GovSpend, we developed a staged analytical framework that standardizes units, extracts package quantities from short item descriptions, groups similar purchases, measures price dispersion, and creates a prioritized review queue. Rule-based parsing was evaluated using a manually labeled sample of 300 transactions. Learned pack-size models and alternative grouping strategies were also evaluated, with emphasis on performance for previously unseen products.*

Results: *The implemented pipeline processed 917,321 records, resolved 43,130 records through the normalization process, and declined 2,865 cases that lacked sufficient evidence for a reliable conversion. The overall coefficient of variation decreased from 0.653 to 0.078, and the same downward pattern was observed across all 58 qualifying product groups. Description-based grouping also produced more internally consistent comparison groups than the supplied category field. However, incomplete package information, limited manual labels, and weaker model performance on unseen products continue to restrict coverage and interpretation.*

Conclusion: *The results indicate that procurement anomaly detection should begin with defensible product and quantity comparisons. The proposed framework can support review prioritization, but its outputs should not be interpreted as proof of overpayment or improper purchasing without additional context and human review.*

***Index Terms*—public procurement, price benchmarking, anomaly detection, unit normalization, natural language processing, data quality**

I. Introduction

Public agencies purchase a broad range of goods and services, often through decentralized systems that record transactions with inconsistent descriptions, categories, quantities, and units of measure. These records can reveal meaningful differences in purchasing behavior, but they can also produce misleading comparisons. A high recorded price may reflect an unusual transaction, a different package size, a specialized specification, a data-entry error, or a legitimate local market condition. For this reason, price anomaly detection in procurement data is not only a statistical problem. It is first a problem of establishing whether two purchases are sufficiently comparable.

This capstone project is sponsored by GovSpend, a government procurement intelligence platform. The project uses a curated, de-identified subset of public purchasing records containing structured fields such as price, quantity, vendor, agency, date, category, and unit of measure, along with short and often highly abbreviated item descriptions. The sponsor has emphasized analytical insight and methodological transparency rather than the delivery of a production-ready application.

A. Background and Motivation

Price transparency can help procurement teams understand market conditions, identify transactions that merit review, and support more informed purchasing decisions. However, useful benchmarks require consistent definitions. A case of gloves priced at one hundred twenty dollars cannot be directly compared with a box of gloves priced at twelve dollars unless the number of items in each package is known. Similar issues arise when agencies use different abbreviations, when categories are assigned incorrectly, or when descriptions omit product specifications. Without normalization and careful grouping, an anomaly model may simply rediscover data-quality problems.

B. Project Goal and Research Questions

The goal of the project is to develop and evaluate a defensible analytical framework for benchmarking prices and identifying potential anomalies in public sector procurement data. The framework is intended to support both methodological learning and practical use by GovSpend and its customers. It focuses on five related questions:

- How much price variability exists among purchases that can reasonably be treated as comparable?
- How effectively can structured fields and short item descriptions be used to normalize quantities and group similar purchases?
- Which statistical or machine learning methods can identify unusual prices without being dominated by heavily skewed distributions?
- How can the analysis distinguish potential pricing anomalies from errors, missing information, or legitimate differences in the underlying purchase?
- Under what conditions could the resulting benchmarks and flags be useful to procurement professionals, and when is human review required?

C. Scope and Expected Contribution

The project is exploratory and methodological. It does not attempt to prove fraud, determine whether an agency overpaid, or automate final purchasing decisions. Instead, it seeks to document a repeatable process for improving comparability, measuring price dispersion, surfacing transactions that warrant further review, and explaining the assumptions behind those results. The expected contribution is a practical assessment of what can be learned from public procurement data, what remains uncertain, and how a combination of automated analysis and human judgment can be used responsibly.

II. Data and Data Preparation

The primary dataset used in this project consists of publicly available Virginia government procurement records obtained through GovSpend. [1] Procurement data was provided as multiple export files, with each row representing an individual line item associated with a purchase order. The dataset contains both purchase order information and item-level details, including the purchasing agency, vendor, purchase description, item description, item category, quantity, unit price, unit of measure (UOM), and total price. Because each purchase order may contain multiple line items, the analysis was performed at the line-item level to enable more detailed comparisons between similar purchases.

The individual export files were merged into a single dataset after filtering to retain only records relevant to the analysis. Initial preprocessing included removing records and missing values in key fields, converting price fields from text to numeric format, and standardizing unit of measure values where equivalent abbreviations existed (e.g. “EA”, “Each”, and “EACH”). Additional summary datasets were created by grouping transactions according to item category and unit of measure to measure pricing behavior across comparable purchases.

One of the most significant challenges encountered during preprocessing was incomplete quantity information. A large proportion of procurement records did not include a recorded unit of measure, making it difficult to determine whether two purchases represented comparable quantities. Even when a quantity was present, it was frequently recorded as one, providing little information about package size or product volume. As a result, two transactions with substantially different prices could represent legitimate differences in packaging rather than pricing inconsistencies. This limitation motivated filtering the analysis to records with valid units of measure whenever possible.

To establish comparable groups of purchases, descriptive statistics were calculated for each Item Category and Unit of Measure combination. These included the number of observations, mean price, median price, minimum and maximum prices, variance, standard deviation, and coefficient of variation (CV). The coefficient of variation was used as a relative measure of price dispersion, allowing categories with different average prices to be compared on a common scale. These aggregated summaries provide the foundation for subsequent anomaly detection methods by identifying categories that exhibit unusually high price variability while also highlighting the limitations imposed by missing or inconsistent procurement information.

III. Methodology

The methodology developed in this project follows a sequential framework designed to improve the comparability of procurement transactions before attempting to identify pricing anomalies. Rather than applying anomaly detection directly to raw procurement data, the proposed approach first establishes a group of purchases that are sufficiently similar to support meaningful price comparisons. This emphasis on comparability is intended to reduce false positives caused by inconsistent procurement records rather than genuine pricing irregularities.

A. Pipeline Overview

Figure 1 summarizes the implemented workflow from the raw transaction extract through the final review queue. The pipeline begins by removing non-product records and separating transactions according to whether they are already priced per item, cannot be converted, or require pack-size interpretation. Rule-based parsing is applied first so that conversions are supported by explicit evidence in the item description. Rows that remain unresolved are passed to the learned model, after which normalized records are grouped by description similarity and evaluated for unusual per-each prices. This sequence is important because it establishes comparability before any transaction is treated as a potential anomaly.

The first stage of the methodology evaluates the quality of the available procurement information. Although the procurement records contain structured fields such as item category, quantity, unit of measure, vendor, and price, many transactions contain missing values or inconsistent formatting. Because quantity and packaging information are critical when comparing prices, records with incomplete units of measure information are separated from those that can reasonably be standardized. Monetary values are converted to numeric data types, and unit of measure abbreviations are standardized to improve consistency across agencies. These preprocessing steps establish a common representation before additional analysis is performed.

The second stage focuses on constructing comparable purchase groups. Multiple grouping strategies were evaluated to determine the appropriate balance between specificity and statistical reliability. The initial approach grouped purchases according to the United Nations Standard Products and Services Code (UNSPC) hierarchy together with standardized units of measure. While this strategy produced highly specific product groupings, many contained relatively few observations. Broader groupings based on item category and unit of measure were also evaluated to increase sample sizes while maintaining reasonable product similarity. These alternative grouping strategies provide a foundation for measuring price

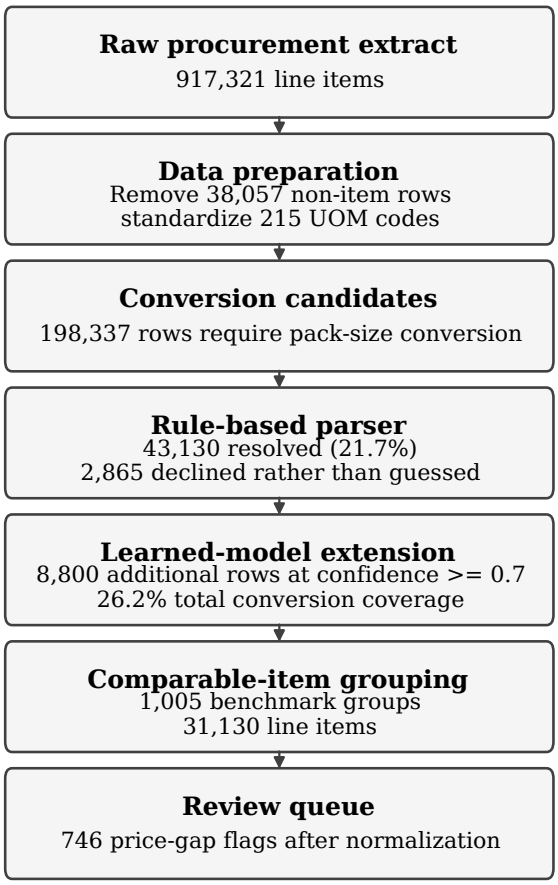


Fig. 1. End-to-end quantity-normalization and price-benchmarking workflow. Counts are taken from the final pipeline run.

dispersion and determining which level of aggregation produces the most meaningful comparisons.

Once comparable groups were established, descriptive statistics were calculated for each group. The number of observations, mean price, median price, minimum price, maximum price, variance, standard deviation, and coefficient of variation (CV) were computed for every grouping. Because procurement prices span several orders of magnitude, the CV was selected as the primary measure of relative price dispersion. Unlike standard deviation alone, CV normalizes variability by the average price, which allows for comparisons between low-cost and high-cost products on a common scale. Groups containing a larger number of observations were prioritized because they provide more reliable estimates of expected pricing behavior and reduce the influence of isolated transactions.

Although structured product classifications provide an initial method for grouping purchases, they are not sufficient to normalize all transactions. Product descriptions frequently contain abbreviations, package counts, and quantity information that are not represented in structured fields. To address this limitation, the methodology incorporates a rule-based parsing approach that extracts package quantities from free-text item descriptions using regular expressions and pattern matching. When packaging information can be identified with sufficient confidence, purchase prices are normalized to an estimated per-unit basis. Transactions that cannot be normalized remain available for analysis but are interpreted with additional caution because their prices may reflect packaging differences rather than true pricing variation.

Finally, the effectiveness of the normalization process is evaluated by comparing price dispersion before and after normalization. Changes in the coefficient of variation across comparable product groups provide an indication of whether the proposed methodology successfully reduces artificial variability introduced by inconsistent procurement records. Rather than producing a final automated anomaly detector, the methodology is intended to establish statistically defensible price benchmarks that procurement professionals can use to prioritize transactions for further review. The resulting framework combines structured procurement data, rule-based normalization, and descriptive statistical analysis to improve transparency while recognizing that human judgment remains an essential component of procurement oversight.

IV. Results

Table 1: Raw Data Metrics

Table 2: Finalized Metrics

Metric	Value
Total Rows Loaded	917,321
Resolved Rows	43,130
Declined Rows	2,865
Coefficient of Variance (Raw)	0.653
Coefficient of Variance (Normalized)	0.078

Table 3: Hand-Labeled Set Metrics

Metric	Value
Total Rows	300
Number Provided	252
Marked Unknowable	46
High Precision	102
Medium Precision	34
Unverifiable Rows	14
Decline Quality	150
Total Disagreements between Parser and Labeler	2

A. *Effect of Normalization on Price Variability*

The finalized normalization process loaded 917,321 procurement records. Of the records evaluated for conversion, 43,130 were resolved and 2,865 were declined because the available structured fields or item descriptions did not support a sufficiently reliable interpretation. The coefficient of variation decreased from 0.653 in the raw data to 0.078 after normalization.

Figure 2 provides a group-level view of this change rather than relying only on the overall summary statistic. Each line

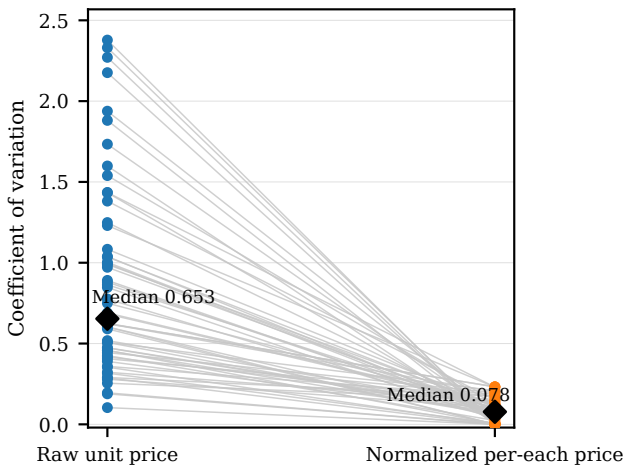


Fig. 2. Coefficient of variation before and after normalization for 58 identical-product groups with multiple observed pack sizes. Each gray line represents one product group; diamonds mark the medians.

connects the raw and normalized coefficient of variation for one of 58 identical- description product groups containing multiple observed pack sizes. Every line moves downward after conversion to a per-each price, showing that the reduction in variability was consistent across all qualifying groups. The figure therefore supports the conclusion that much of the original price spread reflected packaging and unit differences rather than true pricing variation.

A hand-labeled gold set of 300 transactions was used to evaluate the quality of the parser's decisions. Of these records, 252 contained enough information for a labeler to provide a determination and 46 were marked unknowable. The evaluation identified 102 high-precision cases, 34 medium-precision cases, 14 unverifiable rows, and 150 cases in which declining to normalize was considered the appropriate outcome. Only two disagreements were recorded between the parser and the labeler. The gold set was included because aggregate changes in price dispersion alone cannot establish whether individual parsing decisions were correct; a manually reviewed sample provides a direct check on both resolved and declined cases.

B. Learned Model Evaluation

The rule-resolved rows were also used as weak supervision for a learned pack-size model. To avoid an optimistic evaluation in which the same product could appear in both training and testing data, the primary model comparison used a product-level split so that every test product was unseen during training. LinearSVC produced the highest accuracy at 70.3%, followed closely by the modified-huber SGD classifier at 69.3%. Logistic regression achieved 60.6%, ComplementNB achieved 48.2%, and the majority-class baseline achieved only 11.8%. A random row split produced 88.6% accuracy, but the 28-point gap between random and product-level splits demonstrates how strongly repeated products can inflate conventional validation results.

C. Grouping and Outlier-Queue Effects

Description-based grouping was evaluated because the supplied category field often combined unrelated items. Category-plus-UOM groups had a median within-group CV of 0.897, compared with 0.124 at a description- similarity threshold of 0.7 and 0.043 for exact-description groups. The 0.7 threshold formed 1,465 groups with at least five rows and provided a useful compromise between group coverage and within-group price consistency.

Normalization also changed the composition of the review queue rather than merely rescaling prices. In the end-to-end benchmark groups, the normalized approach flagged 746 transactions and the raw-price approach flagged 1,037, but only 73 appeared in both sets. A total of 673 transactions were identified only after normalization, while 964 raw- price flags disappeared after comparability was improved. These results reinforce that anomaly flags should be treated as a prioritized review queue rather than as evidence of overpayment.

V. Discussion

On a practical level, the methodology developed over the course of this project provides GovSpend with a preliminary basis for price and product analysis. One potential bottleneck is the use of a hand-labeled set against which results are compared. Because this set requires analysts to review and label records manually, it creates a material time cost and may also require continued adjustment as end users identify new classification patterns. Results should therefore be reviewed for accuracy, and classifications should be revised when necessary.

The experimental results suggest that the proposed normalization methodology improves the comparability of procurement transactions. The reduction in the coefficient of variation from 0.653 before normalization to 0.078 after normalization indicates that a substantial portion of the observed price variability can be explained by differences in packaging, unit representation, or inconsistent procurement descriptions rather than true pricing differences. This finding supports the hypothesis that normalization should be performed before attempting statistical anomaly detection.

The methodology was intentionally designed to favor precision over coverage. Records were normalized only when sufficient evidence existed within the structured fields or item descriptions to support a reliable conversion. Transactions that could not be confidently interpreted were declined rather than forced into a normalized representation. While this approach reduces the number of records available for analysis, it also reduces the likelihood of generating misleading price comparisons.

The learned-model and grouping results further clarify where automation is most useful. Product-level validation reduced classifier accuracy relative to a random split, showing that generalization to unseen products remains a meaningful challenge. At the same time, description-based grouping substantially improved within-group consistency relative to category-based grouping, indicating that comparable-item

construction is at least as important as the choice of outlier statistic.

The methodology also demonstrates that statistical benchmarking can serve as an effective first stage of procurement analysis rather than as a standalone anomaly detector. By first grouping comparable purchases and reducing artificial price variation, procurement professionals can focus their attention on transactions that are statistically unusual within an appropriate comparison group. This approach can improve the efficiency of manual review by reducing false positives generated by differences in packaging, units of measure, or inconsistent procurement descriptions.

VI. Conclusion and Recommendations

This project demonstrates that reliable procurement price analysis depends on establishing comparability before attempting to identify anomalies. In the raw data, differences in package size, unit representation, and description quality created substantial apparent price variation, even among purchases that initially appeared similar. After prices were normalized to a more comparable basis, the overall coefficient of variation declined from 0.653 to 0.078. The same general reduction was observed across all 58 qualifying product groups examined, indicating that much of the original dispersion was driven by packaging and unit differences rather than true variation in per-item prices. Normalization also changed the composition of the review queue. Many transactions flagged using raw prices were no longer unusual after conversion, while other transactions became visible only after comparability improved. These findings show that anomaly detection should be treated as the final stage of a broader analytical process rather than as a method applied directly to raw procurement records.

By combining structured fields, conservative rule-based parsing, description-based grouping, descriptive statistics, learned models, and a manually reviewed gold set, we developed a repeatable framework for reducing artificial price variation and prioritizing transactions for further review. The framework does not determine whether an agency overpaid or whether a transaction was improper. Instead, it provides GovSpend with a more transparent and defensible way to identify purchases that warrant closer examination while preserving human judgment when the available data are incomplete or ambiguous.

Recommended next steps include developing and continually improving a database of verified transaction information, increasing the size of the hand-labeled evaluation set, and maintaining a separate database of transactions that remain difficult to classify. A verified transaction database would provide future analyses with more reliable quantities and price information from which to estimate comparable prices. Expanding the hand-labeled set would give future analytical models a stronger reference against which to evaluate their own classifications. Finally, logging difficult or unclassified transactions would allow the team to identify recurring patterns that may support additional rules or classifications over time.

Two limitations restricted what we can be claimed from this work. Coverage is partial by design, since 2,865 records were declined where the description supplied a count for a container smaller than the one actually priced. Accepting those counts would have understated the true quantity. The larger constraint is that most item descriptions never state a package size in any recoverable form, which places a ceiling on coverage that better parsing alone cannot manage. The evaluation is also limited because the hand-labeled set comprises 300 transactions annotated by team members rather than procurement experts, and the gap between random-split and product-level accuracy indicates that much of the apparent model performance reflects repeated products rather than learned structure.

Two fields also proved less reliable than expected. The category label is often wrong at the line-item level, so grouping by category mixes unrelated products together. Those groups had a median within-group coefficient of variation of 0.897, compared with 0.124 when items were grouped by how similar their descriptions were. Category codes should, therefore, not be used to decide which purchases are comparable. The unit of measure field has a similar problem, since some records marked as single items are actually priced by the box, with the package count sitting in the description. Future work should teach the parser to read package counts written without a slash, and should train a model to judge whether two similar descriptions really are the same product, because text similarity alone cannot tell an item apart from its accessory. As a note for those continuing the work of this project, Claude was used to assist with the development of the codebase. [2]

VII. Project Contributions

All project members participated in the analysis of the initial raw data, the development of the final paper, and the presentation. Maneesh Malpeddi managed the data-cleaning and exploratory data-analysis process with contributions from Claudio Cela, Jack Hays, and Daniel Ra. Claudio Cela managed the development of the codebase with feedback from Maneesh Malpeddi and Jack Hays.

Acknowledgment

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[2] [3] [1]